Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Sierra Vista Dairy

Physical address of dairy:

22426 E Monte Vista AVE	Denair	Merced	95316
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/01/1978

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X150-X065-XXXX

B. OPERATORS

Burroughs, Bruce

Operator name: Burroughs, Bruce Telephone no.: (209) 874-2528 (Landline) (Cellular)

22312 E Monte Vista AVE	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Burroughs, Bruce

Legal owner name: Burroughs, Bruce Telephone no.: (209) 874-2528 (Landline) (Cellular)

22312 E Monte Vista AVE	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	164	523	356	167	0
Number under roof	1,333	0	0	0	0	0
Maximum number	1,449	179	523	356	167	0
Average number	1,333	164	523	356	167	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 87 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 48,740.71 *tons per reporting period*

Total nitrogen from manure: 603,714.87 *lbs per reporting period*

After ammonia losses (30% loss applied): 422,600.41 *lbs per reporting period*

Total phosphorus from manure: 100,749.50 *lbs per reporting period*

Total potassium from manure: 270,170.08 *lbs per reporting period*

Total salt from manure: 665,354.85 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 18,424,186 *gallons*

Total nitrogen generated: 92,343.56 *lbs*

Total phosphorus generated: 11,509.84 *lbs*

Total potassium generated: 117,049.49 *lbs*

Total salt generated: 589,085.38 *lbs*

	16,533,686	*gallons applied*
+	1,890,500	*gallons exported*
-	0	*gallons imported*
=	18,424,186	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW #11	Ground water
IW #14	Ground water
IW #15	Ground water
IW Jimmy Trees	Ground water
IW New 17	Ground water

Source Description	Type
IW Old 17	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/28/2023	Solid commercial fertilizer *5-11-4*	12.00 *ton*	Dry-weight	0.1	5.000000	11.000000	4.000000	0.000000
08/05/2023	Solid commercial fertilizer *UN32*	18.32 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	12,911.88	2,637.36	959.04	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	12,911.88	2,637.36	959.04	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	Separator solids	8,000.00 *ton*	Dry-weight	69.5		16,600.00	2,100.00	5,090.00		19.30

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
10/12/2023	Process wastewater sludge	1,890,500.00 *gal*	501.00	235.00	0.00	0.00	57.70	742.00		1,660

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	81,008.00	10,248.00	24,839.20	941,840.00
Process wastewater	7,903.89	910.29	11,705.96	26,188.53
Total exports for all materials	88,911.89	11,158.29	36,545.16	968,028.53

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy Trees	75	75	1	manure	X042-X150-X063-XXXX X042-X150-X064-XXXX
Field 10	45	45	2	process wastewater	X042-X150-X060-XXXX X042-X170-X001-XXXX
Field 11	39	39	2	process wastewater	X042-X150-X060-XXXX
Field 13	12	12	2	manure	X042-X150-X066-XXXX
Field 14	54	54	2	process wastewater	X042-X150-X066-XXXX
Field 15	55	55	1	manure	X042-X150-X045-XXXX X042-X150-X066-XXXX
Field 16	11	11	2	manure	X042-X150-X066-XXXX
Field 17	30	30	2	process wastewater	X042-X150-X060-XXXX X042-X150-X064-XXXX
Field 8/9	49	49	2	process wastewater	X042-X170-X001-XXXX
Monica Trees	113	113	1	manure	X042-X150-X064-XXXX X042-X150-X065-XXXX X042-X170-X001-XXXX
Pasture	37	37	0	none	X042-X150-X064-XXXX
Totals for areas that were used for application	483	483	17		
Totals for areas that were not used for application	37	37	0		
Land application area totals	520	520	17		

B. CROPS AND HARVESTS

Dairy Trees

Field name: Dairy Trees

Dairy Trees

04/05/2017: Almond, in shell

Crop: Almond, in shell Acres planted: 75 Plant date: 04/05/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	409.00 *ton*	As-is		36.6	8,880.00	1,360.00	13,200.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	5.45	96.85	14.83	143.97	532.44

Field 10

Field name: Field 10

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	642.00 *ton*	As-is		70.8	4,810.00	1,330.00	10,800.00		9.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.27	137.25	37.95	308.16	798.18

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	1,141.00 *ton*	As-is		65.8	3,900.00	590.00	5,670.00		3.91

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	25.36	197.77	29.92	287.53	678.12

Field 11

Field name: Field 11

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	534.00 *ton*	As-is		75.0	5,380.00	1,330.00	8,360.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.69	147.33	36.42	228.94	595.62

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	1,087.00 *ton*	As-is		67.6	3,760.00	610.00	6,190.00		2.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.87	209.60	34.00	345.05	521.96

Field 13

Field name: Field 13

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	178.00 *ton*	As-is		62.8	5,780.00	1,430.00	11,800.00		7.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.83	171.47	42.42	350.07	840.94

Field 13

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	340.00 *ton*	As-is		67.4	3,720.00	500.00	4,610.00		2.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	28.33	210.80	28.33	261.23	443.36

Field 14

Field name: Field 14

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 54 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	810.00 *ton*	As-is		73.6	5,270.00	700.00	5,980.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	158.10	21.00	179.40	720.72

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	1,947.00 *ton*	As-is		71.2	3,630.00	630.00	4,820.00		33.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	36.06	261.76	45.43	347.58	6,894.98

Field 15

Field name: Field 15

01/17/2020: Almond, in shell

Crop: Almond, in shell Acres planted: 55 Plant date: 01/17/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	144.00 *ton*	As-is		12.3	18,900.00	1,830.00	18,000.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.50	107.50	18.25	116.25	0.00
Total actual harvest content	2.62	98.97	9.58	94.25	473.01

Field 16

Field name: Field 16

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	103.00 *ton*	As-is		68.0	6,520.00	1,310.00	8,830.00		8.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	9.36	122.10	24.53	165.36	484.21

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	269.00 *ton*	As-is		64.9	4,250.00	570.00	5,490.00		5.87

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	24.45	207.86	27.88	268.51	1,007.71

Field 17

Field name: Field 17

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	446.00 *ton*	As-is		61.2	3,690.00	1,200.00	9,070.00		7.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.87	109.72	35.68	269.68	812.17

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	989.00 *ton*	As-is		65.1	3,590.00	700.00	5,060.00		4.41

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	32.97	236.70	46.15	333.62	1,014.77

Field 8/9

Field name: Field 8/9

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	775.00 *ton*	As-is		74.6	5,220.00	700.00	6,170.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.82	165.12	22.14	195.17	674.91

Field 8/9

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	1,452.00 *ton*	As-is		67.7	3,680.00	780.00	5,680.00		5.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	29.63	218.10	46.23	336.63	1,137.08

Monica Trees

Field name: Monica Trees

01/11/2014: Almond, in shell

Crop: Almond, in shell Acres planted: 113 Plant date: 01/11/2014

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	523.00 *ton*	As-is		12.3	18,200.00	1,900.00	16,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	4.63	168.47	17.59	149.96	852.40

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy Trees - 04/05/2017: Almond, in shell

Field name: Dairy Trees

Crop: Almond, in shell

Plant date: 04/05/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	77.93	12.91	46.44	1,511.42	180.00 *ton*
Application event totals		77.93	12.91	46.44	1,511.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW New 17	Ground water	1.84	0.00	0.00	158.75	11,323,692.00 *gal*
Application event totals		1.84	0.00	0.00	158.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	30.08	0.00	0.00	0.00	
Application event totals		30.08	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW New 17	Ground water	1.84	0.00	0.00	158.75	11,323,692.00 *gal*
Application event totals		1.84	0.00	0.00	158.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW New 17	Ground water	1.84	0.00	0.00	158.75	11,323,692.00 *gal*
Application event totals		1.84	0.00	0.00	158.75	

Dairy Trees - 04/05/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW New 17	Ground water	1.84	0.00	0.00	158.75	11,323,692.00 *gal*
Application event totals		1.84	0.00	0.00	158.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW New 17	Ground water	1.84	0.00	0.00	158.75	11,323,692.00 *gal*
Application event totals		1.84	0.00	0.00	158.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW New 17	Ground water	0.92	0.00	0.00	79.38	5,661,846.00 *gal*
Application event totals		0.92	0.00	0.00	79.38	

Field 10 - 11/03/2022: Oats, silage-soft dough

Field name: Field 10

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.57	19.90	131.93	205.43	865,456.00 *gal*
IW #11	Ground water	12.28	0.00	0.00	304.29	5,658,120.00 *gal*
Application event totals		118.84	19.90	131.93	509.72	

Field 10 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.26	17.01	100.48	577.30	841,369.00 *gal*
IW #11	Ground water	11.33	0.00	0.00	280.88	5,222,880.00 *gal*
Application event totals		68.59	17.01	100.48	858.18	

Field 10 - 06/24/2023: Corn, silage

Field name: Field 10

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	12.59	0.00	0.00	312.09	5,803,200.00 *gal*
Application event totals		12.59	0.00	0.00	312.09	

06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	11.65	0.00	0.00	288.68	5,367,960.00 *gal*
Application event totals		11.65	0.00	0.00	288.68	

Field 10 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.00	5.29	117.70	869.04	822,146.00 *gal*
IW #11	Ground water	11.33	0.00	0.00	280.88	5,222,880.00 *gal*
Application event totals		113.33	5.29	117.70	1,149.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	11.02	0.00	0.00	273.08	5,077,800.00 *gal*
Application event totals		11.02	0.00	0.00	273.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	10.39	0.00	0.00	257.47	4,787,640.00 *gal*
Application event totals		10.39	0.00	0.00	257.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.49	4.64	103.26	762.45	721,310.00 *gal*
IW #11	Ground water	10.23	0.00	0.00	253.57	4,715,100.00 *gal*
Application event totals		99.72	4.64	103.26	1,016.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	9.60	0.00	0.00	237.96	4,424,840.00 *gal*
Application event totals		9.60	0.00	0.00	237.96	

Field 11 - 11/03/2022: Oats, silage-soft dough

Field name: Field 11

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Field 11 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.20	18.52	122.80	191.22	698,169.00 *gal*
IW #11	Ground water	14.53	0.00	0.00	360.10	5,803,200.00 *gal*
Application event totals		113.72	18.52	122.80	551.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.13	16.08	94.98	545.70	689,269.00 *gal*
IW #11	Ground water	12.89	0.00	0.00	319.59	5,150,340.00 *gal*
Application event totals		67.02	16.08	94.98	865.29	

Field 11 - 05/28/2023: Corn, silage

Field name: Field 11

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.16	6.28	113.79	773.91	632,315.00 *gal*
IW #11	Ground water	14.17	0.00	0.00	351.10	5,658,120.00 *gal*
Application event totals		121.32	6.28	113.79	1,125.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

Field 11 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	13.08	0.00	0.00	324.09	5,222,880.00 *gal*
Application event totals		13.08	0.00	0.00	324.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	11.99	0.00	0.00	297.09	4,787,640.00 *gal*
Application event totals		11.99	0.00	0.00	297.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.83	4.50	100.20	739.84	606,598.00 *gal*
IW #11	Ground water	11.80	0.00	0.00	292.58	4,715,100.00 *gal*
Application event totals		98.64	4.50	100.20	1,032.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	11.99	0.00	0.00	297.09	4,787,640.00 *gal*
Application event totals		11.99	0.00	0.00	297.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	11.62	0.00	0.00	288.08	4,642,560.00 *gal*
Application event totals		11.62	0.00	0.00	288.08	

Field 13 - 11/03/2022: Oats, silage-soft dough

Field name: Field 13

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 13 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	162.36	26.90	96.76	3,148.80	60.00 *ton*
Application event totals		162.36	26.90	96.76	3,148.80	

03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	3.28	0.00	0.00	193.54	1,987,920.00 *gal*
Application event totals		3.28	0.00	0.00	193.54	

Field 13 - 05/31/2023: Corn, silage

Field name: Field 13

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	3.43	0.00	0.00	202.34	2,078,280.00 *gal*
Application event totals		3.43	0.00	0.00	202.34	

05/31/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	2.98	0.00	0.00	175.95	1,807,200.00 *gal*
Application event totals		2.98	0.00	0.00	175.95	

Field 13 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	112.00	0.00	0.00	0.00	
Application event totals		112.00	0.00	0.00	0.00	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	2.53	0.00	0.00	149.55	1,536,120.00 *gal*
Application event totals		2.53	0.00	0.00	149.55	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	2.68	0.00	0.00	158.35	1,626,480.00 *gal*
Application event totals		2.68	0.00	0.00	158.35	

08/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	112.00	0.00	0.00	0.00	
Application event totals		112.00	0.00	0.00	0.00	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	2.68	0.00	0.00	158.35	1,626,480.00 *gal*
Application event totals		2.68	0.00	0.00	158.35	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	2.38	0.00	0.00	140.76	1,445,760.00 *gal*
Application event totals		2.38	0.00	0.00	140.76	

Field 14 - 11/03/2022: Oats, silage-soft dough

Field 14 - 11/03/2022: Oats, silage-soft dough

Field name: Field 14

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.24	24.32	161.23	251.07	1,269,266.00 *gal*
IW #14	Ground water	3.38	0.00	0.00	199.41	9,216,720.00 *gal*
Application event totals		133.62	24.32	161.23	450.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.82	16.58	97.95	562.76	984,216.00 *gal*
IW #14	Ground water	3.18	0.00	0.00	187.68	8,674,560.00 *gal*
Application event totals		59.00	16.58	97.95	750.44	

Field 14 - 05/31/2023: Corn, silage

Field name: Field 14

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	167.67	9.82	178.05	1,210.98	1,369,966.00 *gal*
IW #14	Ground water	3.51	0.00	0.00	207.23	9,578,160.00 *gal*
Application event totals		171.18	9.82	178.05	1,418.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

Field 14 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	3.44	0.00	0.00	203.32	9,397,440.00 *gal*
Application event totals		3.44	0.00	0.00	203.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	3.34	0.00	0.00	197.45	9,126,360.00 *gal*
Application event totals		3.34	0.00	0.00	197.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	127.44	6.61	147.06	1,085.83	1,232,696.00 *gal*
IW #14	Ground water	3.24	0.00	0.00	191.59	8,855,280.00 *gal*
Application event totals		130.69	6.61	147.06	1,277.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	3.18	0.00	0.00	187.68	8,674,560.00 *gal*
Application event totals		3.18	0.00	0.00	187.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #14	Ground water	3.18	0.00	0.00	187.68	8,674,560.00 *gal*
Application event totals		3.18	0.00	0.00	187.68	

Field 15 - 01/17/2020: Almond, in shell

Field name: Field 15

Crop: Almond, in shell

Plant date: 01/17/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 15 - 01/17/2020: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	76.75	12.71	45.74	1,488.52	130.00 *ton*
Application event totals		76.75	12.71	45.74	1,488.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.52	0.00	0.00	204.23	10,516,129.00 *gal*
Application event totals		2.52	0.00	0.00	204.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	16.64	0.00	0.00	0.00	
Application event totals		16.64	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.52	0.00	0.00	204.23	10,516,129.00 *gal*
Application event totals		2.52	0.00	0.00	204.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.52	0.00	0.00	204.23	10,516,129.00 *gal*
Application event totals		2.52	0.00	0.00	204.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.52	0.00	0.00	204.23	10,516,129.00 *gal*
Application event totals		2.52	0.00	0.00	204.23	

Field 15 - 01/17/2020: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.52	0.00	0.00	204.23	10,516,129.00 *gal*
Application event totals		2.52	0.00	0.00	204.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	1.26	0.00	0.00	102.12	5,258,064.00 *gal*
Application event totals		1.26	0.00	0.00	102.12	

Field 16 - 11/04/2022: Oats, silage-soft dough

Field name: Field 16

Crop: Oats, silage-soft dough Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	147.60	24.45	87.96	2,862.55	50.00 *ton*
Application event totals		147.60	24.45	87.96	2,862.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.38	0.00	0.00	193.06	1,988,160.00 *gal*
Application event totals		2.38	0.00	0.00	193.06	

Field 16 - 06/28/2023: Corn, silage

Field name: Field 16

Crop: Corn, silage Plant date: 06/28/2023

Field 16 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.51	0.00	0.00	203.22	2,092,800.00 *gal*
Application event totals		2.51	0.00	0.00	203.22	

06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.13	0.00	0.00	172.74	1,778,880.00 *gal*
Application event totals		2.13	0.00	0.00	172.74	

07/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	112.00	0.00	0.00	0.00	
Application event totals		112.00	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	2.01	0.00	0.00	162.58	1,674,240.00 *gal*
Application event totals		2.01	0.00	0.00	162.58	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	1.88	0.00	0.00	152.42	1,569,600.00 *gal*
Application event totals		1.88	0.00	0.00	152.42	

Field 16 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	112.00	0.00	0.00	0.00	
Application event totals		112.00	0.00	0.00	0.00	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	1.63	0.00	0.00	132.09	1,360,320.00 *gal*
Application event totals		1.63	0.00	0.00	132.09	

09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	1.51	0.00	0.00	121.93	1,255,680.00 *gal*
Application event totals		1.51	0.00	0.00	121.93	

10/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #15	Ground water	1.51	0.00	0.00	121.93	1,255,680.00 *gal*
Application event totals		1.51	0.00	0.00	121.93	

Field 17 - 11/04/2022: Oats, silage-soft dough

Field name: Field 17

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.13	16.08	106.62	166.03	466,295.00 *gal*
IW Old 17	Ground water	12.33	0.00	0.00	383.82	4,524,000.00 *gal*
Application event totals		98.46	16.08	106.62	549.85	

Field 17 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.51	12.33	72.84	418.48	406,599.00 *gal*
IW Old 17	Ground water	11.06	0.00	0.00	344.11	4,056,000.00 *gal*
Application event totals		52.57	12.33	72.84	762.59	

Field 17 - 06/28/2023: Corn, silage

Field name: Field 17

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Old 17	Ground water	11.91	0.00	0.00	370.58	4,368,000.00 *gal*
Application event totals		11.91	0.00	0.00	370.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.82	5.85	130.19	961.26	606,262.00 *gal*
IW Old 17	Ground water	10.84	0.00	0.00	337.50	3,978,000.00 *gal*
Application event totals		123.67	5.85	130.19	1,298.76	

Field 17 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Old 17	Ground water	10.63	0.00	0.00	330.88	3,900,000.00 *gal*
Application event totals		10.63	0.00	0.00	330.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.32	5.67	126.15	931.44	587,454.00 *gal*
IW Old 17	Ground water	10.21	0.00	0.00	317.64	3,744,000.00 *gal*
Application event totals		119.53	5.67	126.15	1,249.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Old 17	Ground water	9.78	0.00	0.00	304.41	3,588,000.00 *gal*
Application event totals		9.78	0.00	0.00	304.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Old 17	Ground water	9.36	0.00	0.00	291.17	3,432,000.00 *gal*
Application event totals		9.36	0.00	0.00	291.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Old 17	Ground water	9.36	0.00	0.00	291.17	3,432,000.00 *gal*
Application event totals		9.36	0.00	0.00	291.17	

Field 8/9 - 11/04/2022: Oats, silage-soft dough

Field name: Field 8/9

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 8/9 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.57	21.58	143.07	222.79	1,021,996.00 *gal*
IW #11	Ground water	12.72	0.00	0.00	315.27	6,383,520.00 *gal*
Application event totals		128.29	21.58	143.07	538.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.95	17.51	103.44	594.32	943,169.00 *gal*
IW #11	Ground water	12.14	0.00	0.00	300.94	6,093,360.00 *gal*
Application event totals		71.09	17.51	103.44	895.27	

Field 8/9 - 06/28/2023: Corn, silage

Field name: Field 8/9

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	12.00	0.00	0.00	297.36	6,020,820.00 *gal*
Application event totals		12.00	0.00	0.00	297.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	5.00	11.00	4.00	0.00	
Application event totals		5.00	11.00	4.00	0.00	

Field 8/9 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	10.99	0.00	0.00	272.28	5,513,040.00 *gal*
Application event totals		10.99	0.00	0.00	272.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.01	5.39	120.02	886.19	912,896.00 *gal*
IW #11	Ground water	10.84	0.00	0.00	268.70	5,440,500.00 *gal*
Application event totals		114.85	5.39	120.02	1,154.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	10.41	0.00	0.00	257.95	5,222,880.00 *gal*
Application event totals		10.41	0.00	0.00	257.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	10.12	0.00	0.00	250.79	5,077,800.00 *gal*
Application event totals		10.12	0.00	0.00	250.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.56	5.06	112.58	831.19	856,239.00 *gal*
IW #11	Ground water	10.26	0.00	0.00	254.37	5,150,340.00 *gal*
Application event totals		107.82	5.06	112.58	1,085.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #11	Ground water	9.68	0.00	0.00	240.04	4,860,180.00 *gal*
Application event totals		9.68	0.00	0.00	240.04	

Monica Trees - 01/11/2014: Almond, in shell

Field name: Monica Trees

Crop: Almond, in shell

Plant date: 01/11/2014

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	77.59	12.85	46.24	1,504.74	270.00 *ton*
Application event totals		77.59	12.85	46.24	1,504.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Jimmy Trees	Ground water	1.88	0.00	0.00	155.57	18,159,812.00 *gal*
Application event totals		1.88	0.00	0.00	155.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	30.08	0.00	0.00	0.00	
Application event totals		30.08	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Jimmy Trees	Ground water	1.88	0.00	0.00	155.57	18,159,812.00 *gal*
Application event totals		1.88	0.00	0.00	155.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Jimmy Trees	Ground water	1.88	0.00	0.00	155.57	18,159,812.00 *gal*
Application event totals		1.88	0.00	0.00	155.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Jimmy Trees	Ground water	1.88	0.00	0.00	155.57	18,159,812.00 *gal*
Application event totals		1.88	0.00	0.00	155.57	

Monica Trees - 01/11/2014: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Jimmy Trees	Ground water	1.88	0.00	0.00	155.57	18,159,812.00 *gal*
Application event totals		1.88	0.00	0.00	155.57	

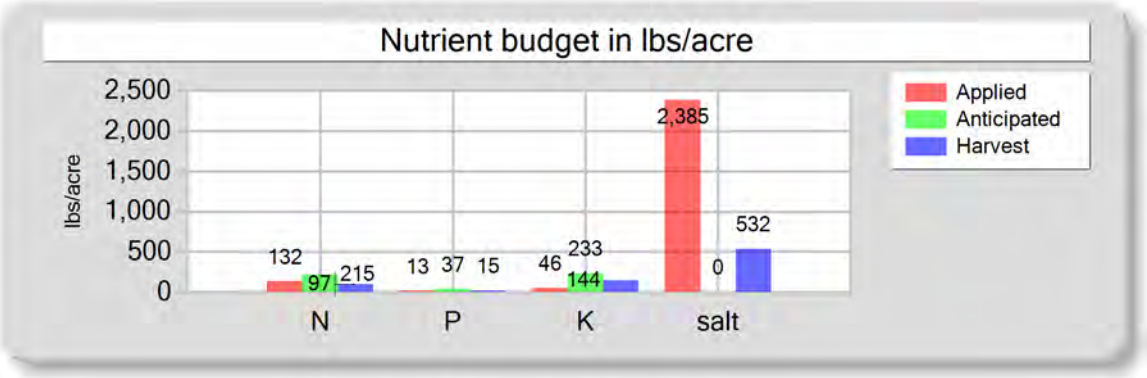
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Jimmy Trees	Ground water	0.94	0.00	0.00	77.78	9,079,906.00 *gal*
Application event totals		0.94	0.00	0.00	77.78	

B. NUTRIENT BUDGET

Dairy Trees - 04/05/2017: Almond, in shell

Field name: Dairy Trees Crop: Almond, in shell Plant date: 04/05/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	30.08	0.00	0.00	0.00
Dry manure	77.93	12.91	46.44	1,511.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.12	0.00	0.00	873.14
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	132.13	12.91	46.44	2,384.57
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	96.85	14.83	143.97	532.44
Nutrient balance	35.28	-1.92	-97.52	1,852.13
Applied to removed ratio	1.36	0.87	0.32	4.48

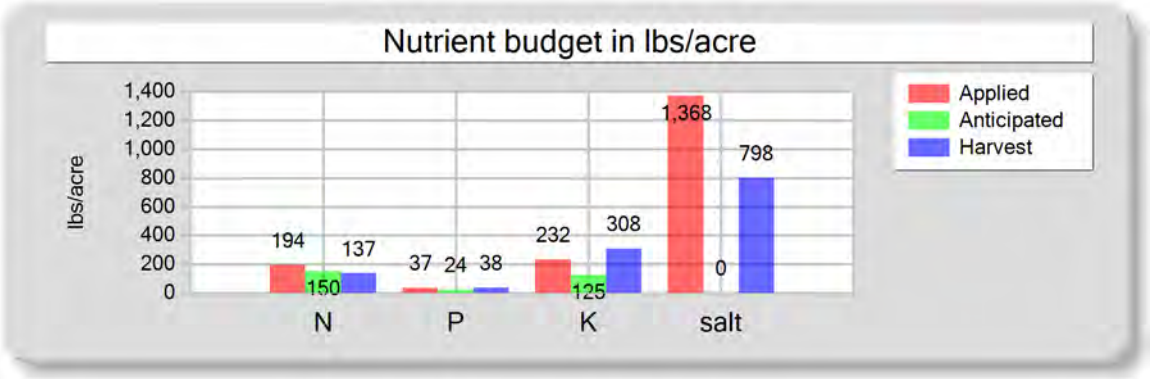
Fresh water applied	
62,280,306.00	*gallons*
2,293.57	*acre-inches*
30.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 10 - 11/03/2022: Oats, silage-soft dough

Field name: Field 10 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.83	36.91	232.41	782.73
Fresh water	23.61	0.00	0.00	585.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.44	36.91	232.41	1,367.90
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	137.25	37.95	308.16	798.18
Nutrient balance	57.19	-1.04	-75.75	569.72
Applied to removed ratio	1.42	0.97	0.75	1.71

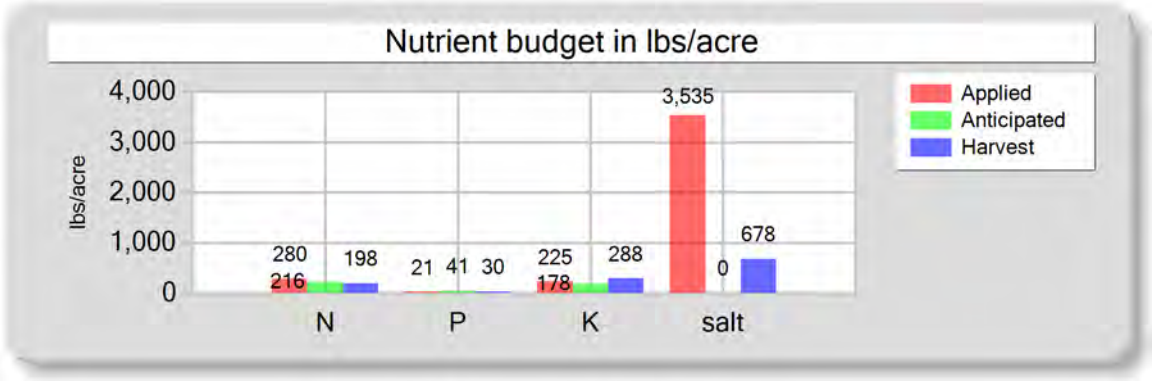
Fresh water applied	
10,881,000.00	*gallons*
400.71	*acre-inches*
8.90	*inches/acre*

Process wastewater applied	
1,706,825.00	*gallons*
62.86	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 10 - 06/24/2023: Corn, silage

Field name: Field 10 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.48	9.93	220.97	1,631.48
Fresh water	76.81	0.00	0.00	1,903.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.29	20.93	224.97	3,535.23
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	197.77	29.92	287.53	678.12
Nutrient balance	82.52	-8.99	-62.57	2,857.11
Applied to removed ratio	1.42	0.70	0.78	5.21

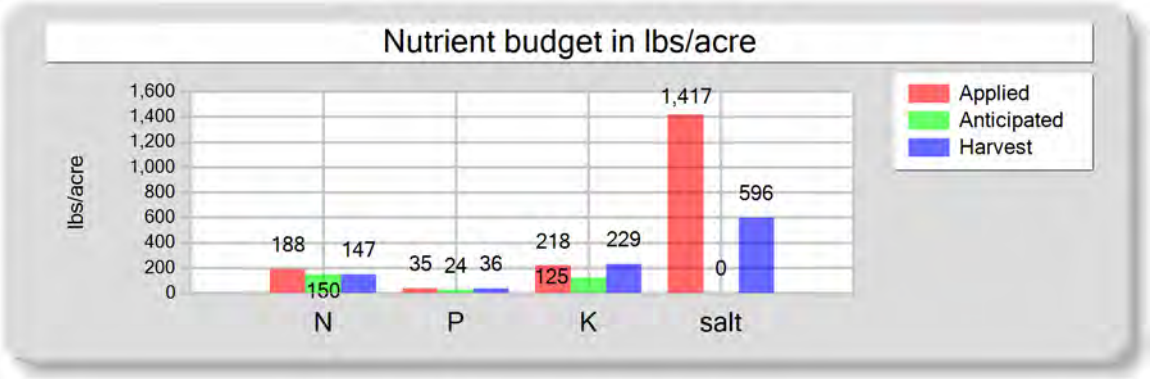
Fresh water applied	
35,399,420.00	*gallons*
1,303.64	*acre-inches*
28.97	*inches/acre*

Process wastewater applied	
1,543,456.00	*gallons*
56.84	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 11/03/2022: Oats, silage-soft dough

Field name: Field 11 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	153.32	34.60	217.78	736.92
Fresh water	27.42	0.00	0.00	679.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.74	34.60	217.78	1,416.61
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	147.33	36.42	228.94	595.62
Nutrient balance	40.42	-1.82	-11.16	821.00
Applied to removed ratio	1.27	0.95	0.95	2.38

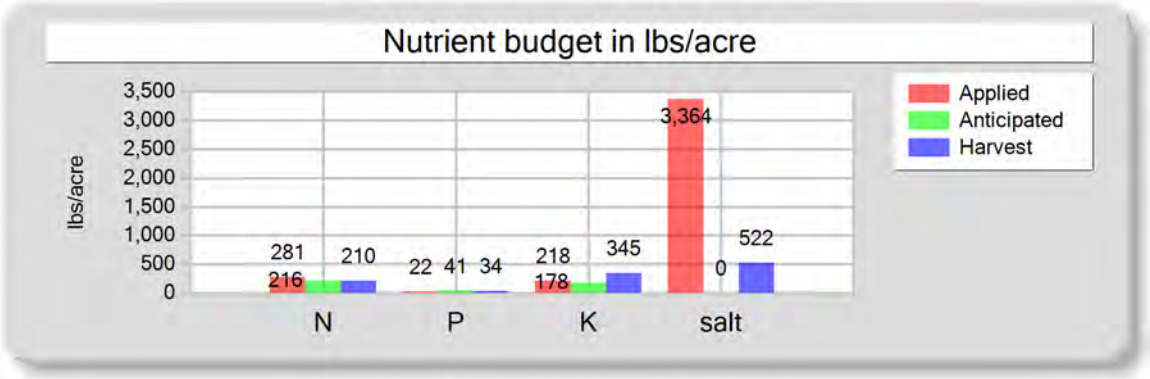
Fresh water applied	
10,953,540.00	*gallons*
403.38	*acre-inches*
10.34	*inches/acre*

Process wastewater applied	
1,387,438.00	*gallons*
51.09	*acre-inches*
1.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 05/28/2023: Corn, silage

Field name: Field 11 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.99	10.78	213.99	1,513.75
Fresh water	74.64	0.00	0.00	1,850.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.63	21.78	217.99	3,363.78
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	209.60	34.00	345.05	521.96
Nutrient balance	71.03	-12.22	-127.06	2,841.82
Applied to removed ratio	1.34	0.64	0.63	6.44

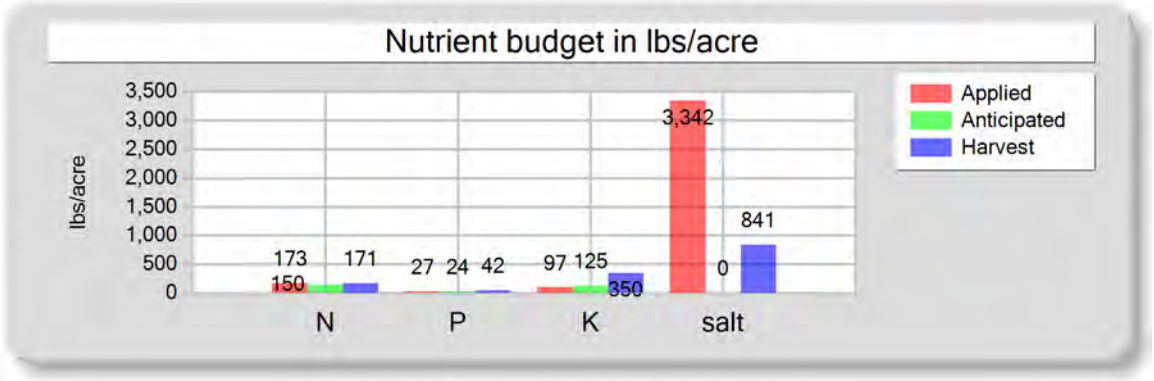
Fresh water applied	
29,813,940.00	*gallons*
1,097.95	*acre-inches*
28.15	*inches/acre*

Process wastewater applied	
1,238,913.00	*gallons*
45.62	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 13 - 11/03/2022: Oats, silage-soft dough

Field name: Field 13 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	162.36	26.90	96.76	3,148.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.28	0.00	0.00	193.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.64	26.90	96.76	3,342.34
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	171.47	42.42	350.07	840.94
Nutrient balance	1.16	-15.53	-253.31	2,501.40
Applied to removed ratio	1.01	0.63	0.28	3.97

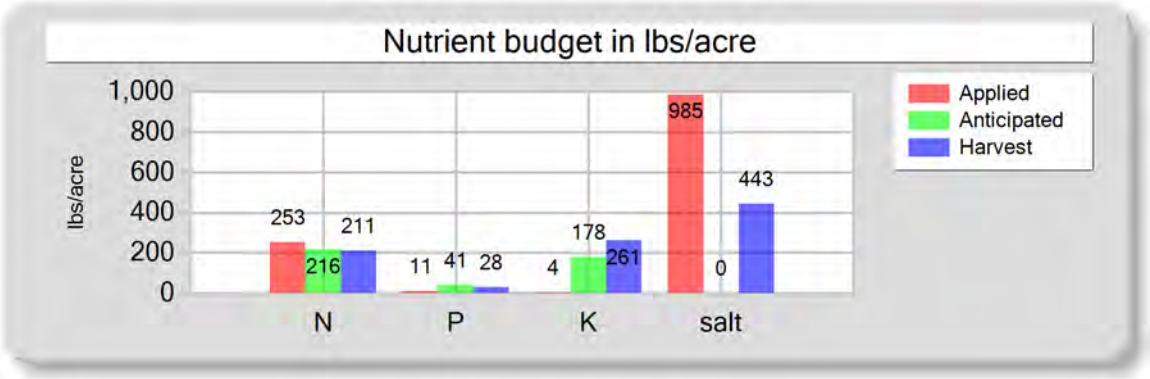
Fresh water applied	
1,987,920.00	*gallons*
73.21	*acre-inches*
6.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 13 - 05/31/2023: Corn, silage

Field name: Field 13 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.68	0.00	0.00	985.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.68	11.00	4.00	985.30
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	210.80	28.33	261.23	443.36
Nutrient balance	41.88	-17.33	-257.23	541.94
Applied to removed ratio	1.20	0.39	0.02	2.22

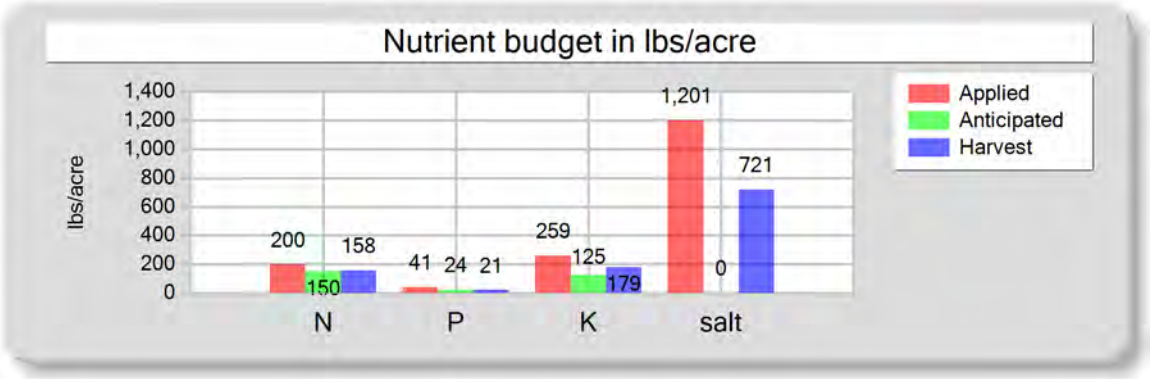
Fresh water applied	
10,120,320.00	*gallons*
372.70	*acre-inches*
31.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 14 - 11/03/2022: Oats, silage-soft dough

Field name: Field 14 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.06	40.90	259.19	813.83
Fresh water	6.55	0.00	0.00	387.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.62	40.90	259.19	1,200.91
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	158.10	21.00	179.40	720.72
Nutrient balance	41.52	19.90	79.79	480.19
Applied to removed ratio	1.26	1.95	1.44	1.67

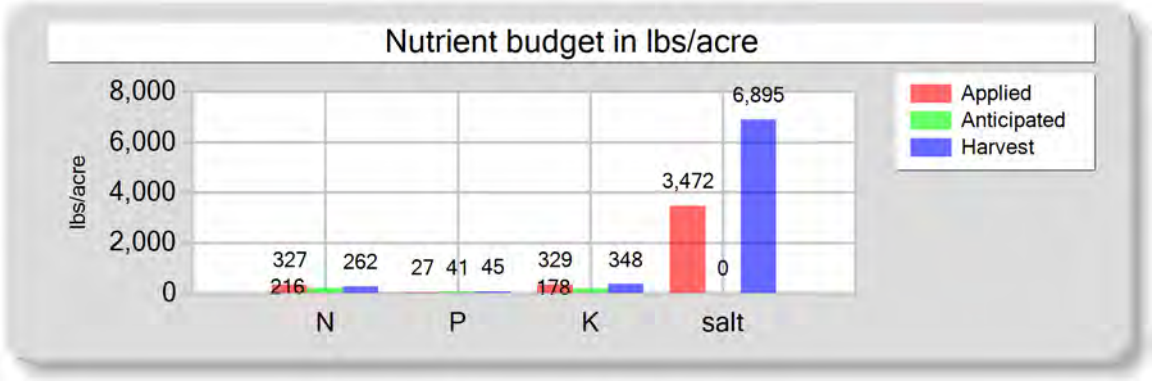
Fresh water applied	
17,891,280.00	*gallons*
658.88	*acre-inches*
12.20	*inches/acre*

Process wastewater applied	
2,253,482.00	*gallons*
82.99	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 14 - 05/31/2023: Corn, silage

Field name: Field 14 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.12	16.43	325.11	2,296.82
Fresh water	19.89	0.00	0.00	1,174.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.01	27.43	329.11	3,471.75
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	261.76	45.43	347.58	6,894.98
Nutrient balance	65.24	-18.00	-18.46	-3,423.23
Applied to removed ratio	1.25	0.60	0.95	0.50

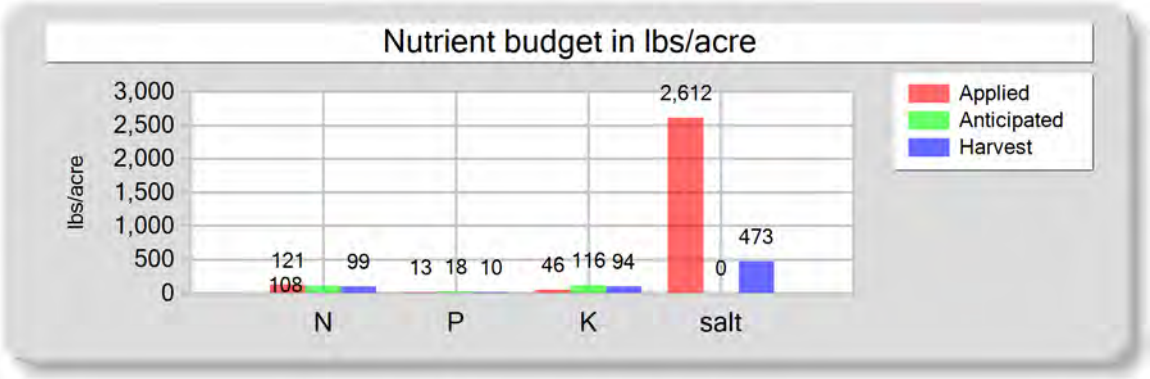
Fresh water applied	
54,306,360.00	*gallons*
1,999.92	*acre-inches*
37.04	*inches/acre*

Process wastewater applied	
2,602,662.00	*gallons*
95.85	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 15 - 01/17/2020: Almond, in shell

Field name: Field 15 Crop: Almond, in shell Plant date: 01/17/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.64	0.00	0.00	0.00
Dry manure	76.75	12.71	45.74	1,488.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.87	0.00	0.00	1,123.29
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	121.26	12.71	45.74	2,611.81
Anticipated crop nutrient removal	107.50	18.25	116.25	0.00
Actual crop nutrient removal	98.97	9.58	94.25	473.01
Nutrient balance	22.29	3.13	-48.51	2,138.81
Applied to removed ratio	1.23	1.33	0.49	5.52

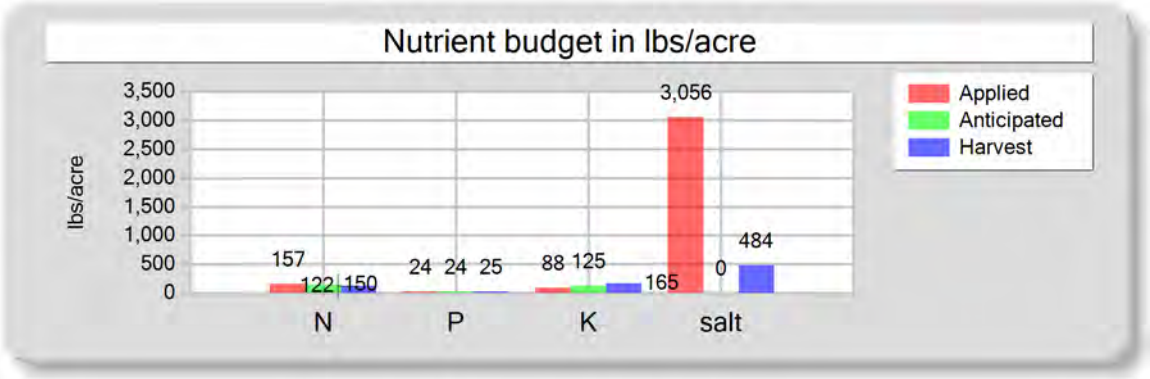
Fresh water applied	
57,838,709.00	*gallons*
2,130.00	*acre-inches*
38.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 16 - 11/04/2022: Oats, silage-soft dough

Field name: Field 16 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	147.60	24.45	87.96	2,862.55
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.38	0.00	0.00	193.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.98	24.45	87.96	3,055.61
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	122.10	24.53	165.36	484.21
Nutrient balance	34.88	-0.08	-77.40	2,571.39
Applied to removed ratio	1.29	1.00	0.53	6.31

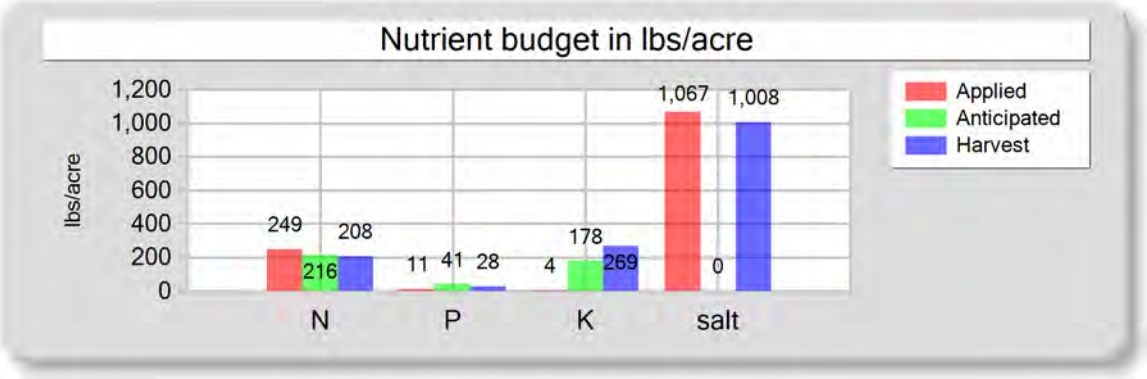
Fresh water applied	
1,988,160.00	*gallons*
73.22	*acre-inches*
6.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 16 - 06/28/2023: Corn, silage

Field name: Field 16 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	229.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.17	0.00	0.00	1,066.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.17	11.00	4.00	1,066.92
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	207.86	27.88	268.51	1,007.71
Nutrient balance	41.31	-16.88	-264.51	59.21
Applied to removed ratio	1.20	0.39	0.01	1.06

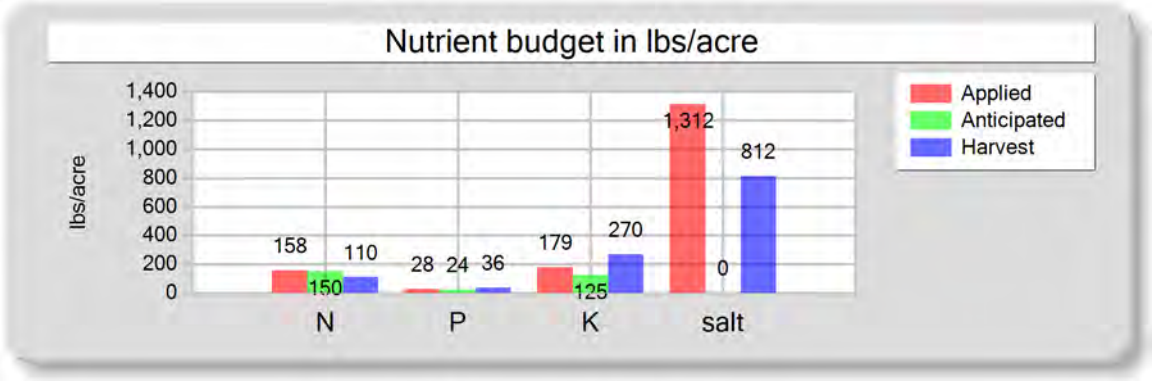
Fresh water applied	
10,987,200.00	*gallons*
404.62	*acre-inches*
36.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 17 - 11/04/2022: Oats, silage-soft dough

Field name: Field 17 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	127.63	28.41	179.46	584.50
Fresh water	23.39	0.00	0.00	727.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.02	28.41	179.46	1,312.44
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	109.72	35.68	269.68	812.17
Nutrient balance	48.31	-7.27	-90.22	500.27
Applied to removed ratio	1.44	0.80	0.67	1.62

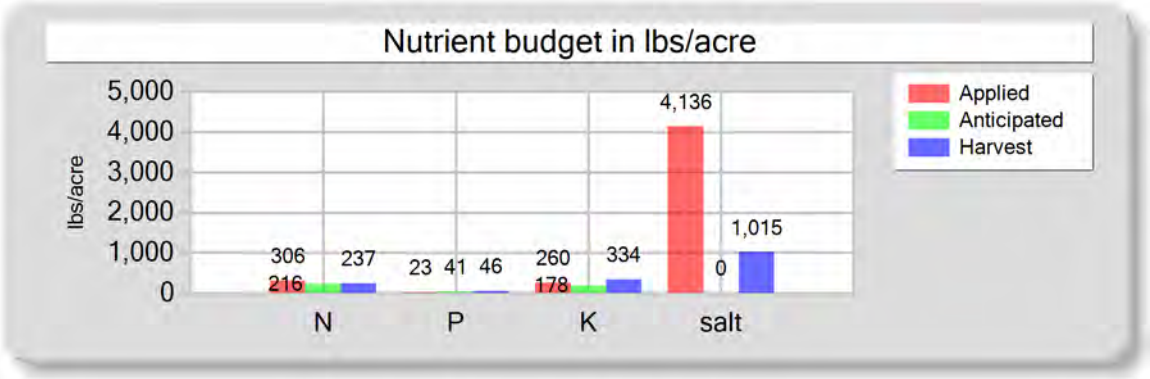
Fresh water applied	
8,580,000.00	*gallons*
315.97	*acre-inches*
10.53	*inches/acre*

Process wastewater applied	
872,894.00	*gallons*
32.15	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 17 - 06/28/2023: Corn, silage

Field name: Field 17 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.14	11.52	256.34	1,892.70
Fresh water	72.08	0.00	0.00	2,243.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.22	22.52	260.34	4,136.06
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	236.70	46.15	333.62	1,014.77
Nutrient balance	69.52	-23.63	-73.28	3,121.28
Applied to removed ratio	1.29	0.49	0.78	4.08

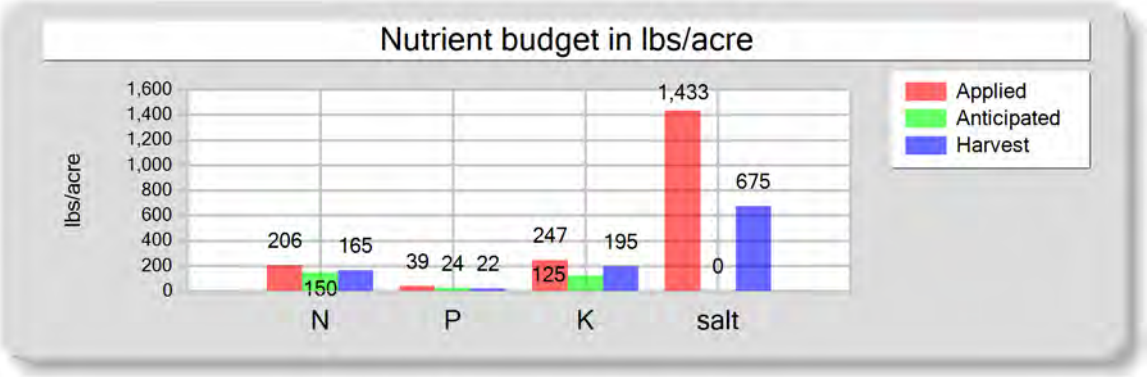
Fresh water applied	
26,442,000.00	*gallons*
973.77	*acre-inches*
32.46	*inches/acre*

Process wastewater applied	
1,193,716.00	*gallons*
43.96	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8/9 - 11/04/2022: Oats, silage-soft dough

Field name: Field 8/9 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.52	39.09	246.51	817.11
Fresh water	24.86	0.00	0.00	616.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.38	39.09	246.51	1,433.33
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	165.12	22.14	195.17	674.91
Nutrient balance	41.26	16.95	51.34	758.41
Applied to removed ratio	1.25	1.77	1.26	2.12

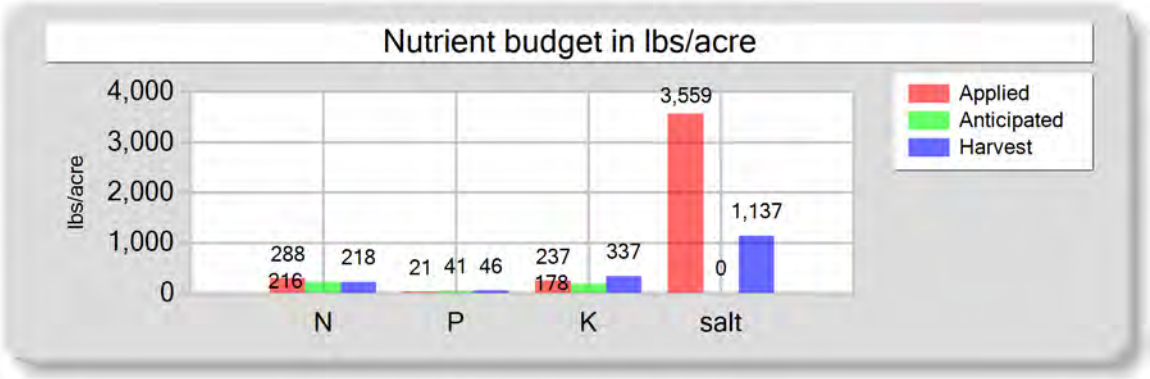
Fresh water applied	
12,476,880.00	*gallons*
459.48	*acre-inches*
9.38	*inches/acre*

Process wastewater applied	
1,965,165.00	*gallons*
72.37	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8/9 - 06/28/2023: Corn, silage

Field name: Field 8/9 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5.00	11.00	4.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.57	10.45	232.60	1,717.38
Fresh water	74.29	0.00	0.00	1,841.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.86	21.45	236.60	3,558.87
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	218.10	46.23	336.63	1,137.08
Nutrient balance	69.76	-24.77	-100.03	2,421.79
Applied to removed ratio	1.32	0.46	0.70	3.13

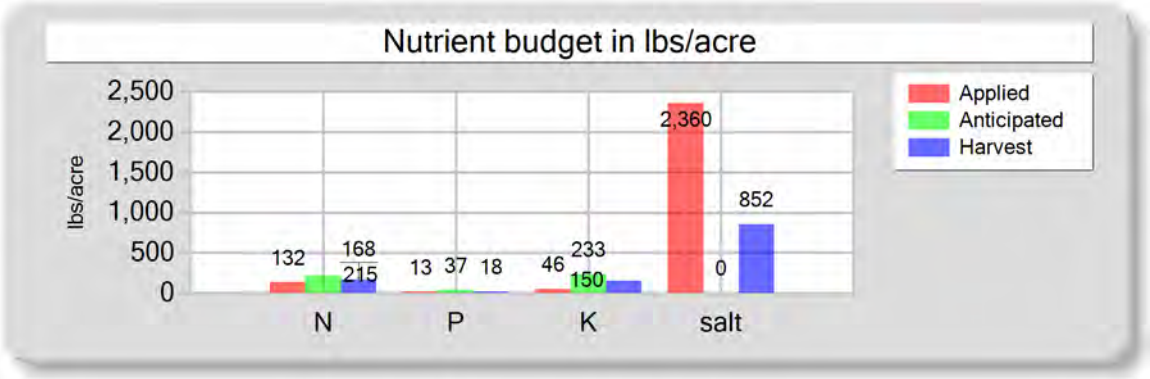
Fresh water applied	
37,285,560.00	*gallons*
1,373.10	*acre-inches*
28.02	*inches/acre*

Process wastewater applied	
1,769,135.00	*gallons*
65.15	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Monica Trees - 01/11/2014: Almond, in shell

Field name: Monica Trees Crop: Almond, in shell Plant date: 01/11/2014



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	30.08	0.00	0.00	0.00
Dry manure	77.59	12.85	46.24	1,504.74
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.33	0.00	0.00	855.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	131.99	12.85	46.24	2,360.35
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	168.47	17.59	149.96	852.40
Nutrient balance	-36.48	-4.73	-103.72	1,507.96
Applied to removed ratio	0.78	0.73	0.31	2.77

Fresh water applied	
99,878,966.00	*gallons*
3,678.20	*acre-inches*
32.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Compost

Sample and source description: Compost

Sample date: 10/13/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	3,280.00	11,800.00	0.00	0.00	0.00	0.00	0.00		38.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,400.00	2,500.00	2,500.00	0.00	0.00	0.00	0.00	0.00		89.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/27/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	1,300.00	1,600.00	0.00	0.00	0.00	0.00	0.00		10.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost

Sample and source description: Compost

Sample date: 10/12/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	7,070.00	28,700.00	0.00	0.00	0.00	0.00	0.00		69.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/12/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	2,100.00	5,090.00	0.00	0.00	0.00	0.00	0.00		19.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	664.00	350.00	0.00	0.00	124.00	822.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,370.00	1,280
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ

Sample and source description: WWQ

Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	367.00	367.00	0.00	0.00	109.00	644.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,360.00	3,700
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	792.00	540.00	0.00	0.00	46.40	841.00	193.00	115.00	284.00	4,340.00	0.00	229.00	168.00	8,150.00	5,720
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	669.00	493.00	0.00	0.00	34.70	772.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,160.00	5,700
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/12/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	235.00	0.00	0.00	57.70	742.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,350.00	1,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	624.00	372.00	0.00	0.00	49.30	706.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7,150.00	5,600
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

IW #11

IW #11

Sample description: IW #11

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.70	0.00	11.20								464.00	290
DL	1.00	0.50	0.10								10.00	10

IW #14

IW #14

Sample description: IW #14

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.37	0.00	2.20								149.00	140
DL	0.50	0.50	0.10								10.00	10

IW #15

IW #15

IW #15

Sample description: IW #15

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.58	0.00	1.40								136.00	128
DL	1.00	0.50	0.10								10.00	10

IW Jimmy Trees

IW Jimmy Trees

Sample description: IW Jimmy Trees

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.40	0.00	1.20								115.00	116
DL	0.50	0.50	0.10								10.00	10

IW New 17

IW New 17

Sample description: IW New 17

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.46	0.00	1.20								116.00	126
DL	0.50	0.50	0.10								10.00	10

IW Old 17

IW Old 17

IW Old 17

Sample description: IW Old 17

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.80	0.00	9.70								490.00	305
DL	0.50	0.50	0.10								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dairy Trees - 04/05/2017: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 36.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,880.00	1,360.00	13,200.00		7.70
DL	100.00	100.00	100.00		0.01

Field 10 - 11/03/2022: Oats, silage-soft dough

Field 10 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,810.00	1,330.00	10,800.00		9.58
DL	100.00	100.00	100.00		0.01

Field 10 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,010.00	660.00	6,510.00		6.49
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	590.00	5,670.00		3.91
DL	100.00	100.00	100.00		0.01

Field 11 - 11/03/2022: Oats, silage-soft dough

Field 11 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,380.00	1,330.00	8,360.00		8.70
DL	100.00	100.00	100.00		0.01

Field 11 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,760.00	610.00	6,190.00		2.89
DL	100.00	100.00	100.00		0.01

Field 13 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,780.00	1,430.00	11,800.00		7.62
DL	100.00	100.00	100.00		0.01

Field 13 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,720.00	500.00	4,610.00		2.40
DL	100.00	100.00	100.00		0.01

Field 14 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,480.00	700.00	5,180.00		9.00
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,270.00	700.00	5,980.00		9.10
DL	100.00	100.00	100.00		0.01

Field 14 - 05/31/2023: Corn, silage

Field 14 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	630.00	4,820.00		33.20
DL	100.00	100.00	100.00		0.01

Field 15 - 01/17/2020: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	1,830.00	18,000.00		10.30
DL	100.00	100.00	100.00		0.01

Field 16 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,520.00	1,310.00	8,830.00		8.08
DL	100.00	100.00	100.00		0.01

Field 16 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,250.00	570.00	5,490.00		5.87
DL	100.00	100.00	100.00		0.01

Field 17 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,690.00	1,200.00	9,070.00		7.04
DL	100.00	100.00	100.00		0.01

Field 17 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	680.00	5,470.00		3.48
DL	100.00	100.00	100.00		0.01

Field 17 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,590.00	700.00	5,060.00		4.41
DL	100.00	100.00	100.00		0.01

Field 8/9 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,880.00	1,170.00	8,630.00		10.70
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,220.00	700.00	6,170.00		8.40
DL	100.00	100.00	100.00		0.01

Field 8/9 - 06/28/2023: Corn, silage

Field 8/9 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,260.00	690.00	5,510.00		3.18
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,680.00	780.00	5,680.00		5.94
DL	100.00	100.00	100.00		0.01

Monica Trees - 01/11/2014: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	1,900.00	16,200.00		10.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

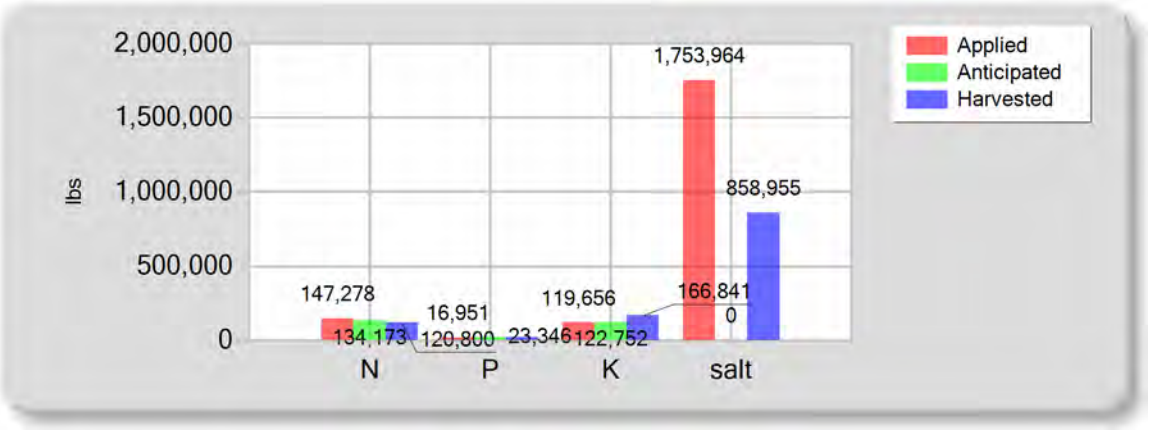
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

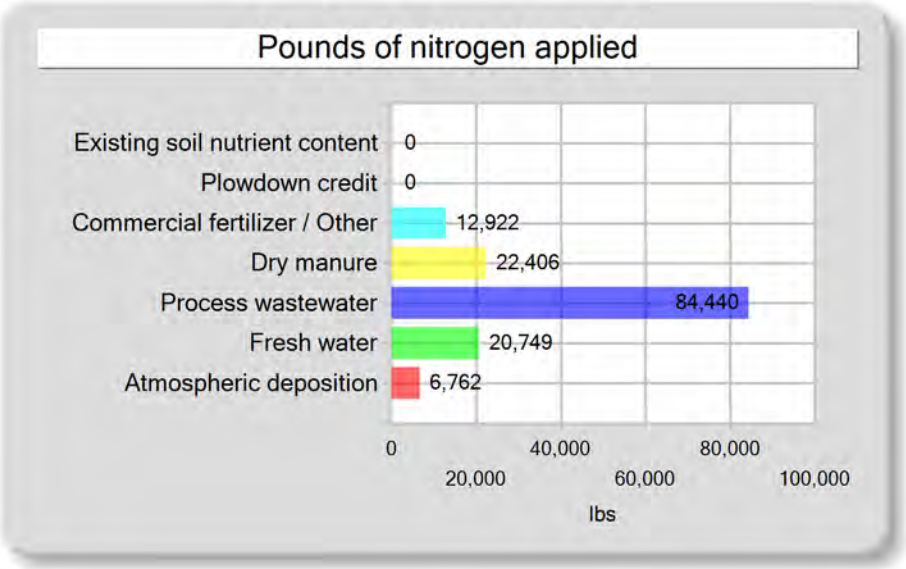
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

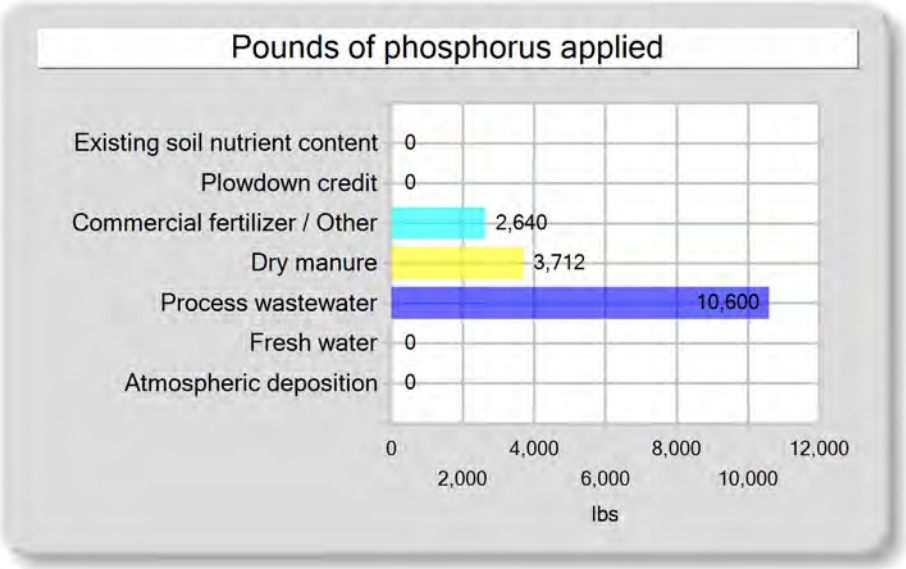
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	12,922.24	2,640.00	960.00	0.00
Dry manure	22,405.68	3,711.65	13,352.88	434,534.40
Process wastewater	84,439.67	10,599.56	105,343.53	562,896.85
Fresh water	20,748.57	0.00	0.00	756,532.76
Atmospheric deposition	6,762.00	0.00	0.00	0.00
Total nutrients applied	147,278.16	16,951.20	119,656.41	1,753,964.01
Anticipated crop nutrient removal	134,172.50	23,345.75	122,751.75	0.00
Actual crop nutrient removal	120,799.90	20,245.54	166,840.70	858,954.52
Nutrient balance	26,478.26	-3,294.34	-47,184.29	895,009.48
Applied to removed ratio	1.22	0.84	0.72	2.04

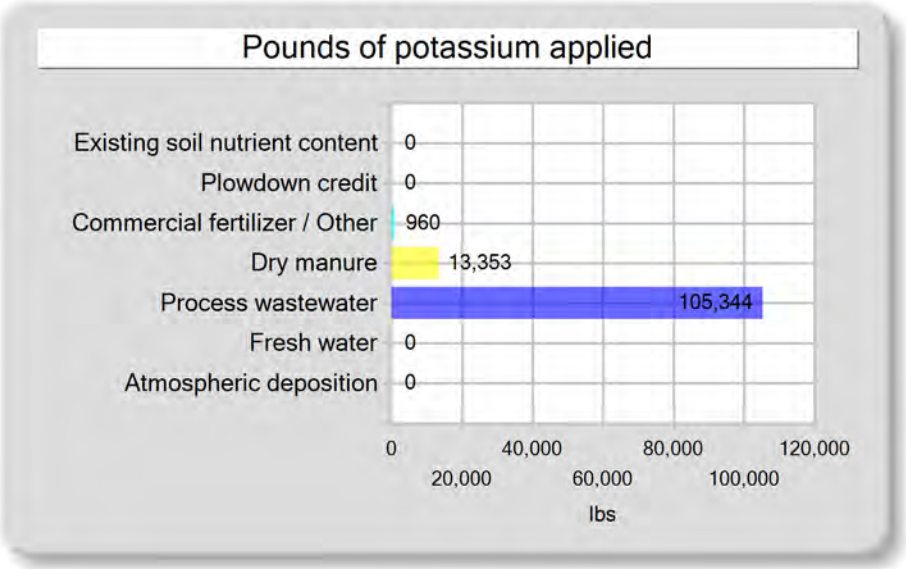
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

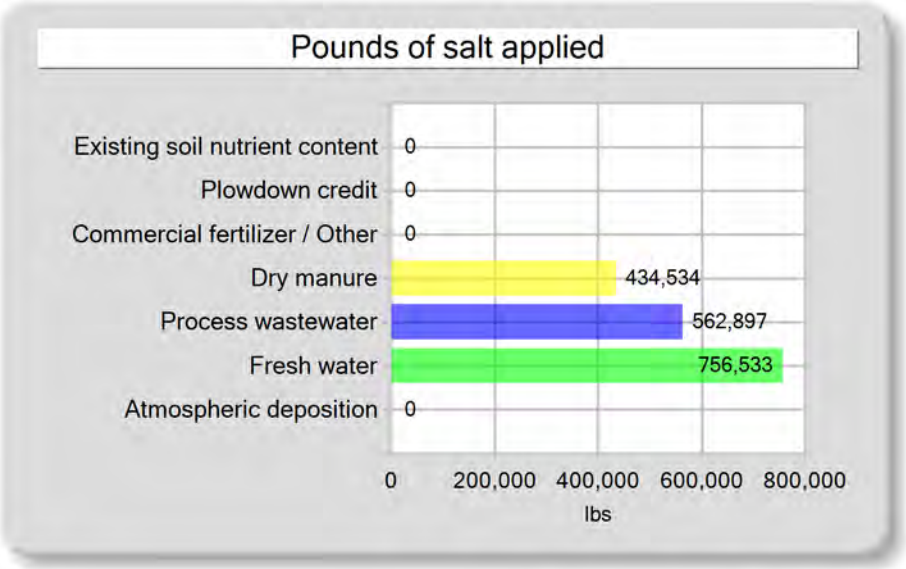


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Mature Cows were removed from the facility in December 2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

James Burroughs (Apr 22, 2024 13:19 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bruce Burroughs	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 22, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: BRUCE BURROUGHS

Name of Dairy Facility: SIERRA VISTA DAIRY

Facility Address: 22426 E. MONTE VISTA AVE. DENAIR 95316
Number and Street City Zip Code

Contact Person Name and Phone Number: JIMMY BURROUGHS 209-613-4184
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: WARD BURROUGHS

Address of Hauling Company /Person: 22470 E. MONTE VISTA DENAIR 95316
Number and Street City Zip Code

Contact Person: WARD BURROUGHS 209-874-1309
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

VISTA LIVESTOCK 22470 E. MONTE VISTA DENAIR 95316 209-874-1309
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

22323 E. MONTE VISTA AVE. DENAIR 95316
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 11/14/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 8000 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 69.5 % MOISTURE
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

320 LOADS @ 25 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: James Burroughs (Apr 22, 2024 13:19 PDT) Date: Apr 22, 2024

Hauler's Signature: Ward Burroughs Date: 5/26/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: BRUCE BURROUGHS

Name of Dairy Facility: SIERRA VISTA DAIRY

Facility Address: 22426 E. MONTE VISTA AVE. DENAIR 95316
Number and Street City Zip Code

Contact Person Name and Phone Number: JIMMY BURROUGHS 209-613-4184
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: F+A LAGOON MANAGEMENT

Address of Hauling Company /Person: 2930 GEER RD PMB 214 TURLOCK 95382
Number and Street City Zip Code

Contact Person: SERAPHIN FONTES 209-678-1257
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

ROBERT REHER 4032 BLEDSOE RD. DENAIR 95316 209-874-2632
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

042-020-025
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 10/12/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Attachment D
Waste Discharge Requirements General Order No. R5-2007-0035
Existing Milk Cow Dairies

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,890,500 Gallons

Method used to determine volume of process wastewater: _____

199 LOADS OF SLURRY @ 9500 GALLONS PER LOAD

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: James Burroughs (Apr 22, 2024 13:19 PDT) Date: Apr 22, 2024

Hauler's Signature: _____ Date: 4-2-24

DELLAVALLE™
LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23J0925-01	Bruce Well	Ground Water	Jimmy		10/11/2023 10:40
23J0925-02	DW #2	Ground Water	Jimmy		10/11/2023 10:47
23J0925-03	DW #3	Ground Water	Jimmy		10/11/2023 10:45
23J0925-04	DW #4 @ Shop	Ground Water	Jimmy		10/11/2023 10:15

Default Cooler Temperature on Receipt °C: -0.1
Containers Intact
COC/Labels Agree

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Sample Results

Sample: Bruce Well
23J0925-01 (Water)

Sampled: 10/11/2023 10:40
Sampled By: Jimmy

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.26**	mmhos/cm	0.01	1		10/13/23 11:52	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**263**	umhos/cm	10.0	1		10/13/23 11:52	SM 2510 B		BEJ0487
Ammonia (as N)	*	mg/L	0.00	1		10/11/23 10:40	Field		BEJ0564
Nitrate Nitrogen as NO3N	**6.6**	mg/L	0.1	1	10	10/13/23 03:59	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:52	SM 4500-H+	H	BEJ0487
pH	**7.9**	units	1.0	1		10/13/23 11:52	SM 4500-H+	H	BEJ0487
Total Nitrogen	**6.63**	mg/L	1.00	1		10/19/23 09:37	SM 4500-NH3 C		BEJ0642

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DELLAVALLE™ LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Sample Results
(Continued)

Sample: DW #2
23J0925-02 (Water)

Sampled: 10/11/2023 10:47
Sampled By: Jimmy

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.12**	mmhos/cm	0.01	1		10/13/23 11:54	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**124**	umhos/cm	10.0	1		10/13/23 11:54	SM 2510 B		BEJ0487
Ammonia (as N)	*	mg/L	0.00	1		10/11/23 10:47	Field		BEJ0564
Nitrate Nitrogen as NO3N	**1.5**	mg/L	0.1	1	10	10/13/23 04:19	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:54	SM 4500-H+	H	BEJ0487
pH	**7.9**	units	1.0	1		10/13/23 11:54	SM 4500-H+	H	BEJ0487
Total Nitrogen	**1.51**	mg/L	1.00	1		10/19/23 09:38	SM 4500-NH3 C		BEJ0642

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DELLAVALLE™ LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Sample Results
(Continued)

Sample: DW #3
23J0925-03 (Water)

Sampled: 10/11/2023 10:45
Sampled By: Jimmy

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.12**	mmhos/cm	0.01	1		10/13/23 11:55	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**124**	umhos/cm	10.0	1		10/13/23 11:55	SM 2510 B		BEJ0487
Ammonia (as N)	*	mg/L	0.00	1		10/11/23 10:45	Field		BEJ0564
Nitrate Nitrogen as NO3N	**1.5**	mg/L	0.1	1	10	10/13/23 04:39	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:55	SM 4500-H+	H	BEJ0487
pH	**7.9**	units	1.0	1		10/13/23 11:55	SM 4500-H+	H	BEJ0487
Total Nitrogen	**1.52**	mg/L	1.00	1		10/19/23 09:40	SM 4500-NH3 C		BEJ0642

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Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Sample Results
(Continued)

Sample: DW #4 @ Shop
23J0925-04 (Water)

Sampled: 10/11/2023 10:15
Sampled By: Jimmy

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.14**	mmhos/cm	0.01	1		10/13/23 11:56	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**142**	umhos/cm	10.0	1		10/13/23 11:56	SM 2510 B		BEJ0487
Ammonia (as N)	*	mg/L	0.00	1		10/11/23 10:15	Field		BEJ0564
Nitrate Nitrogen as NO3N	**2.2**	mg/L	0.1	1	10	10/13/23 05:00	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:56	SM 4500-H+	H	BEJ0487
pH	**7.9**	units	1.0	1		10/13/23 11:56	SM 4500-H+	H	BEJ0487
Total Nitrogen	**2.18**	mg/L	1.00	1		10/19/23 09:41	SM 4500-NH3 C		BEJ0642

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Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0487										
Blank (BEJ0487-BLK1)					Prepared & Analyzed: 10/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.3		1.0	units						
Blank (BEJ0487-BLK2)					Prepared & Analyzed: 10/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.8		1.0	units						
Blank (BEJ0487-BLK3)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.8		1.0	units						
Duplicate (BEJ0487-DUP1)		**Source: 23J0836-01**			Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	0.73		0.01	mmhos/cm		0.75			2.62	10
pH	7.3		1.0	units		7.3			0.136	10
Electrical Conductivity umhos	730		10.0	umhos/cm		749			2.62	10
Duplicate (BEJ0487-DUP2)		**Source: 23J0923-04**			Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	0.48		0.01	mmhos/cm		0.49			1.31	10
pH	7.7		1.0	units		7.6			0.524	10
Electrical Conductivity umhos	484		10.0	umhos/cm		490			1.31	10
Reference (BEJ0487-SRM1)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	519			umhos/cm	538.0		96.4	90-110		
Reference (BEJ0487-SRM2)					Prepared & Analyzed: 10/13/2023					
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEJ0487-SRM3)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	970			umhos/cm	1000		97.0	90-110		
Electrical Conductivity umhos	970			umhos/cm	1000		97.0	90-110		
Reference (BEJ0487-SRM4)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	976			umhos/cm	1000		97.6	90-110		
Electrical Conductivity umhos	976			umhos/cm	1000		97.6	90-110		
Reference (BEJ0487-SRM5)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	984			umhos/cm	1000		98.4	90-110		

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DELLAVALLE™ LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0487 (Continued)										
Reference (BEJ0487-SRM5)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity umhos	984			umhos/cm	1000		98.4	90-110		
Reference (BEJ0487-SRM6)					Prepared & Analyzed: 10/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEJ0487-SRM7)					Prepared & Analyzed: 10/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEJ0487-SRM8)					Prepared & Analyzed: 10/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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DELLAVALLE™ LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0493										
Blank (BEJ0493-BLK1)					Prepared & Analyzed: 10/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEJ0493-BLK2)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEJ0493-BLK3)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEJ0493-BS1)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	5.4		0.1	mg/L	5.000		108	90-110		
LCS (BEJ0493-BS2)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEJ0493-DUP1)		**Source: 23J0923-02**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	2.2		0.1	mg/L		2.2			0.460	10
Duplicate (BEJ0493-DUP2)		**Source: 23J0923-06**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	1.2		0.1	mg/L		1.2			1.21	10
Matrix Spike (BEJ0493-MS1)		**Source: 23J0923-02**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	7.5		0.1	mg/L	5.000	2.2	106	90-110		
Matrix Spike (BEJ0493-MS2)		**Source: 23J0923-06**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	6.5		0.1	mg/L	5.000	1.2	106	90-110		
Reference (BEJ0493-SRM1)					Prepared & Analyzed: 10/12/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEJ0493-SRM2)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEJ0493-SRM3)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:44

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0642										
Blank (BEJ0642-BLK1)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Total Nitrogen	ND		1.00	mg/L						
Blank (BEJ0642-BLK2)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Total Nitrogen	ND		1.00	mg/L						
Blank (BEJ0642-BLK3)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Total Nitrogen	ND		1.00	mg/L						

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10/12/23 15:38 23J0925

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of **Water** Samples 4 No. of Bottles 4

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [X] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** _____ **Tissue** _____ **Manure** _____

Logged By _____ Results Needed By _____

BILL TO: / 09 (Acct#/Cons.) Purchase Order No _____

SIERRA VISTA
Company Name

22426 E. MONTE VISTA AVE.
Address

DENAIR, CA 95316
City, State Zip

Telephone _____ Fax _____

Cell/Email _____

REQUESTED BY F&R Ag Services

PROJECT _____

Crop _____ Intended _____

Sampled By JIMMY

Analysis and Bottles Required:

[X] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[ ] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. BRUCE WELL		10/11 10:40	-0.1
2. DW #2		10:47	0.6
3. DW #3		10:45	0.9
4. DW #4 @ SHOP		10:15	1.3
5			
6.			
7.			
8.			
9.			
10.			

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	[signature]	F+R	10/12 11:00	
Second	[signature]	DLV	10/12-23 1:55	10-12-23 3:20
Third				
Fourth	[signature]	DLL	10/12/23 15:38	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$_____ In
$_____ Out

Amt Paid _____ Rec By _____ Check # _____ Date _____

Signature _____

Paste Lab

10/12/23 15:38 23J0925

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1	1	1						
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

DRAFT
SEND OUT

DELLAVALLE™
LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23J0923-01	IW #11	Canal Water	Matt		10/11/2023 14:00
23J0923-02	IW #14	Canal Water	Matt		10/11/2023 14:15
23J0923-03	IW #15	Canal Water	Matt		10/11/2023 14:30
23J0923-04	IW Old 17	Canal Water	Matt		10/11/2023 14:45
23J0923-05	IW New 17	Canal Water	Matt		10/11/2023 15:00
23J0923-06	IW Jimmy Trees	Canal Water	Matt		10/11/2023 15:15

Default Cooler Temperature on Receipt °C: 1.3
Containers Intact
COC/Labels Agree

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

Sierra Vista Dairy
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Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Sample Results

Sample: IW #11
23J0923-01 (Water)

Sampled: 10/11/2023 14:00
Sampled By: Matt

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.46**	mmhos/cm	0.01	1		10/13/23 11:45	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**464**	umhos/cm	10.0	1		10/13/23 11:45	SM 2510 B		BEJ0487
Nitrate Nitrogen as NO3N	**11.2**	mg/L	0.1	1	10	10/12/23 22:53	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:45	SM 4500-H+	H	BEJ0487
pH	**7.8**	units	1.0	1		10/13/23 11:45	SM 4500-H+	H	BEJ0487
Total Filterable Solids (TDS)	**290**	mg/L	10.0	1		10/17/23 15:22	SM 2540 C		BEJ0580
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/19/23 09:20	SM 4500-NH3 C		BEJ0642
Total Nitrogen	**11.7**	mg/L	1.00	1		10/19/23 09:20	SM 4500-NH3 C		BEJ0642

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Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Sample Results
(Continued)

Sample: IW #14
23J0923-02 (Water)

Sampled: 10/11/2023 14:15
Sampled By: Matt

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.15**	mmhos/cm	0.01	1		10/13/23 11:46	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**149**	umhos/cm	10.0	1		10/13/23 11:46	SM 2510 B		BEJ0487
Nitrate Nitrogen as NO3N	**2.2**	mg/L	0.1	1	10	10/12/23 23:13	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:46	SM 4500-H+	H	BEJ0487
pH	**7.6**	units	1.0	1		10/13/23 11:46	SM 4500-H+	H	BEJ0487
Total Filterable Solids (TDS)	**140**	mg/L	10.0	1		10/17/23 15:22	SM 2540 C		BEJ0580
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/19/23 09:22	SM 4500-NH3 C		BEJ0642
Total Nitrogen	**2.37**	mg/L	1.00	1		10/19/23 09:22	SM 4500-NH3 C		BEJ0642

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DELLAVALLE™
LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Sample Results
(Continued)

Sample: IW #15
23J0923-03 (Water)

Sampled: 10/11/2023 14:30
Sampled By: Matt

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.14**	mmhos/cm	0.01	1		10/13/23 11:47	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**136**	umhos/cm	10.0	1		10/13/23 11:47	SM 2510 B		BEJ0487
Nitrate Nitrogen as NO3N	**1.4**	mg/L	0.1	1	10	10/12/23 23:34	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:47	SM 4500-H+	H	BEJ0487
pH	**7.8**	units	1.0	1		10/13/23 11:47	SM 4500-H+	H	BEJ0487
Total Filterable Solids (TDS)	**128**	mg/L	10.0	1		10/17/23 15:22	SM 2540 C		BEJ0580
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/19/23 09:23	SM 4500-NH3 C		BEJ0642
Total Nitrogen	**1.58**	mg/L	1.00	1		10/19/23 09:23	SM 4500-NH3 C		BEJ0642

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Account Manager: Ben Nydam
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Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Sample Results
(Continued)

Sample: IW Old 17
23J0923-04 (Water)

Sampled: 10/11/2023 14:45
Sampled By: Matt

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.49**	mmhos/cm	0.01	1		10/13/23 11:49	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**490**	umhos/cm	10.0	1		10/13/23 11:49	SM 2510 B		BEJ0487
Nitrate Nitrogen as NO3N	**9.7**	mg/L	0.1	1	10	10/12/23 23:54	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:49	SM 4500-H+	H	BEJ0487
pH	**7.6**	units	1.0	1		10/13/23 11:49	SM 4500-H+	H	BEJ0487
Total Filterable Solids (TDS)	**305**	mg/L	10.0	1		10/17/23 15:22	SM 2540 C		BEJ0580
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/19/23 09:33	SM 4500-NH3 C		BEJ0642
Total Nitrogen	**9.80**	mg/L	1.00	1		10/19/23 09:33	SM 4500-NH3 C		BEJ0642

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DELLAVALLE™
LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Sample Results
(Continued)

Sample: IW New 17
23J0923-05 (Water)

Sampled: 10/11/2023 15:00
Sampled By: Matt

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.12**	mmhos/cm	0.01	1		10/13/23 11:50	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**116**	umhos/cm	10.0	1		10/13/23 11:50	SM 2510 B		BEJ0487
Nitrate Nitrogen as NO3N	**1.2**	mg/L	0.1	1	10	10/13/23 00:14	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:50	SM 4500-H+	H	BEJ0487
pH	**7.9**	units	1.0	1		10/13/23 11:50	SM 4500-H+	H	BEJ0487
Total Filterable Solids (TDS)	**126**	mg/L	10.0	1		10/17/23 15:22	SM 2540 C		BEJ0580
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/19/23 09:34	SM 4500-NH3 C		BEJ0642
Total Nitrogen	**1.46**	mg/L	1.00	1		10/19/23 09:34	SM 4500-NH3 C		BEJ0642

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Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Sample Results
(Continued)

Sample: IW Jimmy Trees
23J0923-06 (Water)

Sampled: 10/11/2023 15:15
Sampled By: Matt

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.12**	mmhos/cm	0.01	1		10/13/23 11:51	SM 2510 B		BEJ0487
Electrical Conductivity umhos	**115**	umhos/cm	10.0	1		10/13/23 11:51	SM 2510 B		BEJ0487
Nitrate Nitrogen as NO3N	**1.2**	mg/L	0.1	1	10	10/13/23 02:58	EPA 300.0		BEJ0493
Temperature	**25.0**	units	0.0	1		10/13/23 11:51	SM 4500-H+	H	BEJ0487
pH	**7.8**	units	1.0	1		10/13/23 11:51	SM 4500-H+	H	BEJ0487
Total Filterable Solids (TDS)	**116**	mg/L	10.0	1		10/17/23 15:22	SM 2540 C		BEJ0580
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/19/23 09:35	SM 4500-NH3 C		BEJ0642
Total Nitrogen	**1.40**	mg/L	1.00	1		10/19/23 09:35	SM 4500-NH3 C		BEJ0642

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DELLAVALLE™ LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0487										
Blank (BEJ0487-BLK1)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEJ0487-BLK2)					Prepared & Analyzed: 10/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.8		1.0	units						
Blank (BEJ0487-BLK3)					Prepared & Analyzed: 10/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEJ0487-DUP1)		**Source: 23J0836-01**			Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	0.73		0.01	mmhos/cm		0.75			2.62	10
pH	7.3		1.0	units		7.3			0.136	10
Electrical Conductivity umhos	730		10.0	umhos/cm		749			2.62	10
Duplicate (BEJ0487-DUP2)		**Source: 23J0923-04**			Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	0.48		0.01	mmhos/cm		0.49			1.31	10
Electrical Conductivity umhos	484		10.0	umhos/cm		490			1.31	10
pH	7.7		1.0	units		7.6			0.524	10
Reference (BEJ0487-SRM1)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	519			umhos/cm	538.0		96.4	90-110		
Reference (BEJ0487-SRM2)					Prepared & Analyzed: 10/13/2023					
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEJ0487-SRM3)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	970			umhos/cm	1000		97.0	90-110		
Electrical Conductivity umhos	970			umhos/cm	1000		97.0	90-110		
Reference (BEJ0487-SRM4)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	976			umhos/cm	1000		97.6	90-110		
Electrical Conductivity umhos	976			umhos/cm	1000		97.6	90-110		
Reference (BEJ0487-SRM5)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity	984			umhos/cm	1000		98.4	90-110		

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DELLAVALLE™ LABORATORY INC

Sierra Vista Dairy
22426 E Monte Vista Avenue
Denair, CA 95316

Account# 00-0015355
Account Manager: Ben Nydam
Submitted By: F & R Ag Services

Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0487 (Continued)										
Reference (BEJ0487-SRM5)					Prepared & Analyzed: 10/13/2023					
Electrical Conductivity umhos	984			umhos/cm	1000		98.4	90-110		
Reference (BEJ0487-SRM6)					Prepared & Analyzed: 10/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEJ0487-SRM7)					Prepared & Analyzed: 10/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEJ0487-SRM8)					Prepared & Analyzed: 10/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0493										
Blank (BEJ0493-BLK1)					Prepared & Analyzed: 10/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEJ0493-BLK2)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEJ0493-BLK3)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEJ0493-BS1)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	5.4		0.1	mg/L	5.000		108	90-110		
LCS (BEJ0493-BS2)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEJ0493-DUP1)		**Source: 23J0923-02**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	2.2		0.1	mg/L		2.2			0.460	10
Duplicate (BEJ0493-DUP2)		**Source: 23J0923-06**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	1.2		0.1	mg/L		1.2			1.21	10
Matrix Spike (BEJ0493-MS1)		**Source: 23J0923-02**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	7.5		0.1	mg/L	5.000	2.2	106	90-110		
Matrix Spike (BEJ0493-MS2)		**Source: 23J0923-06**			Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	6.5		0.1	mg/L	5.000	1.2	106	90-110		
Reference (BEJ0493-SRM1)					Prepared & Analyzed: 10/12/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEJ0493-SRM2)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEJ0493-SRM3)					Prepared: 10/12/2023 Analyzed: 10/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Received: 10/12/2023 15:38
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0580										
Blank (BEJ0580-BLK1)					Prepared: 10/16/2023	Analyzed: 10/17/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEJ0580-BS1)					Prepared: 10/16/2023	Analyzed: 10/17/2023				
Total Filterable Solids (TDS)	25.0		10.0	mg/L	2000		1.25	0-200		
Duplicate (BEJ0580-DUP1)			**Source: 23J0836-01**		Prepared: 10/16/2023	Analyzed: 10/17/2023				
Total Filterable Solids (TDS)	490		10.0	mg/L		490			0.00	10
Duplicate (BEJ0580-DUP2)			**Source: 23J1009-02**		Prepared: 10/16/2023	Analyzed: 10/17/2023				
Total Filterable Solids (TDS)	2920		10.0	mg/L		2880			1.15	10
Reference (BEJ0580-SRM1)					Prepared: 10/16/2023	Analyzed: 10/17/2023				
Total Filterable Solids (TDS)	313			mg/L	325.0		96.4	90-110		

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Received: 10/12/2023 15:38
Reported: 10/20/2023 14:41

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEJ0642										
Blank (BEJ0642-BLK1)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEJ0642-BLK2)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEJ0642-BLK3)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEJ0642-BS1)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	5.19		1.00	mg/L	5.709		90.9	90-110		
LCS (BEJ0642-BS2)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	5.66		1.00	mg/L	5.709		99.1	90-110		
Duplicate (BEJ0642-DUP1)		**Source: 23J0866-02**			Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	270		3.50	mg/L		271			0.175	10
Duplicate (BEJ0642-DUP2)		**Source: 23J0938-01**			Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		0.919				10
Matrix Spike (BEJ0642-MS1)		**Source: 23J0866-02**			Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	263		3.50	mg/L	9.990	271	NR	90-110		
Matrix Spike (BEJ0642-MS2)		**Source: 23J0938-01**			Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	7.50		1.40	mg/L	7.992	0.919	82.3	90-110		
Reference (BEJ0642-SRM1)					Prepared: 10/17/2023	Analyzed: 10/19/2023				
Kjeldahl Nitrogen (TKN), Total	22.2			mg/L	23.80		93.1	90-110		

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10/12/23 15:38 23J0923

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of **Water** Samples 6 No. of Bottles 6

Water Type: [ ] Drinking Water [ ] Wastewater
[✓] Ag Water [ ] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** ____ **Tissue** ____ **Manure** ____

Logged By ____ Results Needed By ____

BILL TO: ____ / 09 ____
Acct#/Cons. Purchase Order No

SIERRA VISTA
Company Name
22426 E. MONTE VISTA AVE.
Address
DENAIR, CA 95316
City, State Zip

Telephone Fax

Cell/Email

REQUESTED BY F&R Ag Services

PROJECT ____

Crop ____ Intended ____

Sampled By MATT

Analysis and Bottles Required:

[ ] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[X] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. IW #11		10/11 2:00	1.3
2. IW #14		2:15	8.2
3. IW #15		2:30	0.8
4. IW OLD 17		2:45	0.4
5. IW NEW 17		3:00	1.7
6. IW JIMMY TREES		10/11 3:15	0.4
7.			
8.			
9.			
10.			

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First		F+R	10/12 11:00	
Second	JLF	DLV	10-12-23 1:55	10-12-23 3:20
Third				
Fourth		DLI	10/12/23 15:38	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24% or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$____ In
$____ Out

Amt Paid ____ Rec By ____ Check # ____ Date ____

Signature ____

Pi

10/12/23 15:38 23J0923

Shipping Information: Shipped In □ Picked-Up ☒ Walk In □ DLI Sampler □ Other □ _____

□ Samples refridgerated before pick up | □ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box □ None □ | **Refrigerant:** Wet Ice ☒ Blue Ice □ None □

Samples Preserved with HNO_3 or H_2SO_4 were: □ Received Preserved ☒ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic	1	1	1	1	1	1				
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic	1	1	1	1	1	1				
Plastics	* pH Value	<2	<2	<2	<2	<2	<2				
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	Other:										
Special	Other:										

pH Strips
Lot: 10BDH4501 Exp: Jan 2025

SENT OUT

Sierra Vista 2023 Annual Report

Final Audit Report 2024-04-22

Created:	2024-04-22
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAYsAZmGQMID--qFHbBh0TVXMqUsEj5be3

"Sierra Vista 2023 Annual Report" History

Document created by Sean Locke (slocke@fragservices.com)
2024-04-22 - 7:14:53 PM GMT

Document emailed to sierravistacows@gmail.com for signature
2024-04-22 - 7:15:55 PM GMT

Email viewed by sierravistacows@gmail.com
2024-04-22 - 7:16:01 PM GMT

Signer sierravistacows@gmail.com entered name at signing as James Burroughs
2024-04-22 - 8:19:07 PM GMT

Document e-signed by James Burroughs (sierravistacows@gmail.com)
Signature Date: 2024-04-22 - 8:19:09 PM GMT - Time Source: server

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